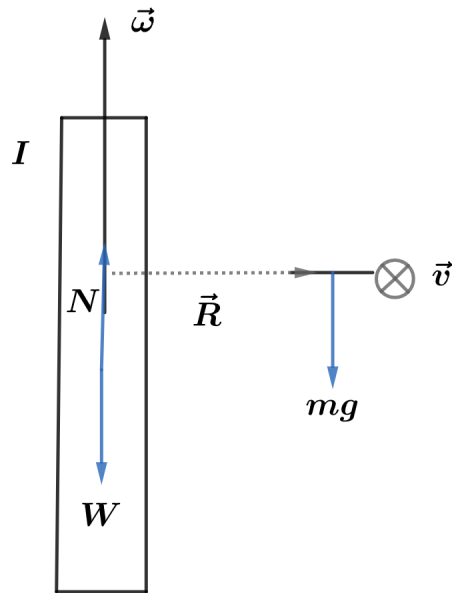


2020A F=ma Exam: Problem 13

Kevin S. Huang



Choosing the rotation axis along the ballerina, we see that the total angular momentum is

$$\vec{L} = I\vec{\omega} + \vec{R} \times m\vec{v}$$

which is completely in the vertical direction. The external torque is in the perpendicular direction so the vertical component of angular momentum stays the same. The answer is C.